

RK-4th order method (Runge-Kutta method)

NA ST
python (P.C)

④ Using RK-4th order method, solve $y' = \ln(x+y)$, $y(0)=2$ to estimate $y(1)$.

solⁿ Here,

$$y' = \ln(x+y), \quad y(0)=2$$

$$\therefore f(x,y) = \ln(x+y)$$

$$\therefore x_0 = 0; y_0 = 2, \quad x_n = 1$$

Take $\therefore h = 0.5$ (say)

The RK-4th order method is given by

$$y_{n+1} = y_n + \frac{h}{6} (m_1 + 2m_2 + 2m_3 + m_4)$$

where,

$$m_1 = f(x_n, y_n)$$

$$m_2 = f(x_n + h/2, y_n + m_1 h/2)$$

$$m_3 = f(x_n + h/2, y_n + m_2 h/2)$$

$$m_4 = f(x_n + h, y_n + m_3 h)$$

$$n = 0, 1, 2, \dots$$

For $n=0$,

$$y_1 = y_0 + \frac{h}{6} (m_1 + 2m_2 + 2m_3 + m_4) \quad \text{--- (1)}$$

where,

$$m_1 = f(0, 2) = 0.6931$$

$$m_2 = f(x_0 + h/2, y_0 + m_1 h/2) = f(0.25, 2.1733) = 0.8851$$

$$m_3 = f(x_0 + h/2, y_0 + m_2 h/2) = f(0.25, 2.2213) = 0.9047$$

$$m_4 = f(x_0 + h, y_0 + m_3 h) = f(0.5, 2.4524) = 1.0826$$

From (1),

$$y_1 = 2 + \frac{0.5}{6} (0.6931 + 2 \times 0.8851 + 2 \times 0.9047 + 1.0826)$$

$$\therefore y_1 = 2.4463 = y(0.5)$$

For $n=2$:

$$y_2 = y_1 + \frac{h}{6} (m_1 + 2m_2 + 2m_3 + m_4) \quad \text{--- (2)}$$

where,

$$m_1 = f(x_1, y_1) = f(0.5, 2.4463) = 1.0805$$

$$m_2 = f(x_1 + h/2, y_1 + m_1 h/2) = f(0.75, 2.7164) = 1.2431$$

$$m_3 = f(x_1 + h/2, y_1 + m_2 h/2) = f(0.75, 2.7571) = 1.2548$$

$$m_4 = f(x_1 + h, y_1 + m_3 h) = f(1, 2.76) = 1.3244$$

From (2),

$$y_2 = 2.4463 + \frac{0.5}{6} (1.0805 + 2 \times 1.2431 + 2 \times 1.2548 + 1.3244)$$

$$\therefore y_2 = 3.0630 \approx y(1) \quad \underline{\text{Ans}}$$

2021-Spring

② Apply RK-4th order method, solve $\frac{dy}{dx} + 2y = x^2, y(1) = 5$.

at $x=1.5$ with $h=0.25$

sol Here,

$$\frac{dy}{dx} + 2y = x^2, y(1) = 5$$

$$\Rightarrow \frac{dy}{dx} = x^2 - 2y, y(1) = 5$$

$$\therefore f(x, y) = x^2 - 2y$$

$$\therefore x_0 = 1, y_0 = 5, x_n = 1.5$$

$$\therefore h = 0.25$$

The RK-4th order method is given by

$$y_{n+1} = y_n + \frac{h}{6} [m_1 + 2m_2 + 2m_3 + m_4]$$

where,

$$m_1 = f(x_n, y_n)$$

$$m_2 = f(x_n + h/2, y_n + m_1 h/2)$$

$$m_3 = f(x_n + h/2, y_n + m_2 h/2)$$

$$m_4 = f(x_n + h, y_n + m_3 h)$$

for $n=0$:

$$y_1 = y_0 + \frac{h}{6} (m_1 + 2m_2 + 2m_3 + m_4) \quad \text{--- (1)}$$

where,

$$m_1 = f(x_0, y_0) = f(1, 5) = -9$$

$$m_2 = f(x_0 + h/2, y_0 + m_1 h/2) = f(1.125, 3.8575) = -3.9875 \\ = -6.4844$$

$$m_3 = f(x_0 + h/2, y_0 + m_2 h/2) = f(1.125, 4.1805) = -7.1134$$

$$m_4 = f(x_0 + h, y_0 + m_3 h) = f(1.25, 3.2217) = -4.8809$$

From (1),

$$y_1 = 5 + 0.25 \frac{(-9 + 2x - 6.4844 + 2x - 7.1134 - 4.8809)}{6}$$

$$\therefore y_1 = 3.2885 = y(1.25) \quad \underline{\text{Ans}}$$

For $n=1$:

$$y_2 = y_1 + \frac{h}{6} (m_1 + 2m_2 + 2m_3 + m_4) \quad \text{--- (2)}$$

where,

$$m_1 = f(x_1, y_1) = f(1.25, 3.2885) = -5.0145$$

$$m_2 = f(x_1 + h/2, y_1 + m_1 h/2) = f(1.375, 2.6617) = -3.4328$$

$$m_3 = f(x_1 + h/2, y_1 + m_2 h/2) = f(1.375, 2.8594) = -3.8282$$

$$m_4 = f(x_1 + h, y_1 + m_3 h) = f(1.5, 2.3315) = -3.6630 \\ = -2.4130$$

From (2),

$$y_2 = 3.2885 + 0.25 \frac{(-5.0145 + 2x - 3.4328 + 2x - 3.8282 - 2.4130)}{6}$$

$$= 2.3739$$

$$\therefore y_2 = 2.3739 = y(1.5) \quad \underline{\text{Ans}}$$

5.a) Using Runge-Kutta method of 4th order, solve $\frac{dy}{dx} = y + \sin x$ with $y(0) = 1$ in the interval $0 \leq x \leq 1$.

Take $h = 0.5$

sol Here,

$$\frac{dy}{dx} = y + \sin x, \quad y(0) = 1$$

$$\therefore x_0 = 0, y_0 = 1, x_n = 1$$

$$\therefore h = 0.5$$

The Runge-Kutta 4th order method is given by

$$y_{n+1} = y_n + \frac{h}{6} (m_1 + 2m_2 + 2m_3 + m_4)$$

where,

$$m_1 = f(x_n, y_n)$$

$$m_2 = f(x_n + h/2, y_n + m_1 h/2)$$

$$m_3 = f(x_n + h/2, y_n + m_2 h/2)$$

$$m_4 = f(x_n + h, y_n + m_3 h)$$

For $n=0$:

$$y_1 = y_0 + \frac{h}{6} (m_1 + 2m_2 + 2m_3 + m_4) \quad \text{--- (1)}$$

where,

$$m_1 = f(x_0, y_0) = f(0, 1) = 1$$

$$m_2 = f(x_0 + h/2, y_0 + m_1 h/2) = f(0.25, 1.25) = 1.4974$$

$$m_3 = f(x_0 + h/2, y_0 + m_2 h/2) = f(0.25, 1.3744) = 1.6218$$

$$m_4 = f(x_0 + h, y_0 + m_3 h) = f(0.5, 1.8109) = 2.2903$$

From (1),

$$y_1 = 1 + \frac{0.5}{6} (1 + 2 \times 1.4974 + 2 \times 1.6218 + 2.2903)$$

$$\therefore y_1 = 1.7941 = y(0.5) \quad \underline{\text{Ans}}$$

For $n=1$:

$$y_2 = y_1 + \frac{h}{6} (m_1 + 2m_2 + 2m_3 + m_4) \quad \text{--- (2)}$$

where,

$$m_1 = f(x_1, y_1) = f(0.5, 1.7941) = 2.2731$$

$$m_2 = f(x_1 + h/2, y_1 + m_1 h/2) = f(0.75, 2.3624) = 3.044$$

$$m_3 = f(x_1 + h/2, y_1 + m_2 h/2) = f(0.75, 2.5551) = 3.2367$$

$$m_4 = f(x_1 + h, y_1 + m_3 h) = f(1, 3.4125) = 4.2540$$

From (2),

$$y_2 = 1.7941 + \frac{0.5}{6} (2.2731 + 2 \times 3.044 + 2 \times 3.2367 + \cancel{4.2540} + 4.2540)$$

$$\therefore y_2 = 3.3848 = y(1) \quad \underline{\text{Ans}}$$

2023-Spring

NAIST
Python (P-C)

6.a) Using the RK-4th order, solve for y at $x=1.2, 1.4$ from
 $\frac{dy}{dx} = \frac{2xy + e^x}{x^2 + xe^x}$ given $x_0 = 1, y_0 = 0$.

So Here,

$$\frac{dy}{dx} = \frac{2xy + e^x}{x^2 + xe^x}, \quad y(1) = 0$$

$$\therefore x_0 = 1, y_0 = 0$$

$$\therefore f(x) = \frac{2xy + e^x}{x^2 + xe^x}$$

$$\therefore h = 0.2$$

The RK-4th order method is given by

$$y_{n+1} = y_n + \frac{h}{6} (m_1 + m_2 + 2m_3 + m_4) \quad \text{--- (A')}$$

where,

$$m_1 = f(x_n, y_n)$$

$$m_2 = f(x_n + h/2, y_1 + m_1 h/2)$$

$$m_3 = f(x_n + h/2, y_1 + m_2 h/2)$$

$$m_4 = f(x_n + h, y_1 + m_3 h)$$

For $n=0$:

$$y_0 = y_0 + \frac{h}{6} (m_1 + m_2 + 2m_3 + m_4) \quad \text{--- (1)}$$

where,

$$m_1 = f(x_0, y_0) = f(1, 0) = 0.7311$$

$$m_2 = f(x_0 + h/2, y_0 + m_1 h/2) = f(1.1, 0.0731) = 0.7011$$

$$m_2 = f(x_0 + h/2, y_0 + m_2 h/2) = f(1.1, 0.0701) = 0.6996$$

$$m_3 = f(x_0 + h, y_0 + m_3 h) = f(1.2, 0.1399) = 0.6740$$

From (1),

$$y_1 = 0 + \frac{0.2}{6} (0.7311 + 2 \times 0.7011 + 2 \times 0.6996 + 0.6740)$$

$$\therefore y_1 = 0.1402 = y(1.2) \quad \underline{\text{Ans}}$$

For $n=2$:

$$y_2 = y_1 + \frac{h}{6} (m_1 + 2m_2 + 2m_3 + m_4) \quad \text{--- (2)}$$

where,

$$m_1 = f(x_1, y_1) = f(1.2, 0.1402) = 0.6741$$

$$m_2 = f(x_1 + h/2, y_1 + m_1 h/2) = f(1.3, 0.2076) = 0.6515$$

$$m_3 = f(x_1 + h/2, y_1 + m_2 h/2) = f(1.3, 0.2054) = 0.6507$$

$$m_4 = f(x_1 + h, y_1 + m_3 h) = f(1.4, 0.2703) = 0.6301$$

From (2),

$$y_2 = 0.1402 + \frac{0.2}{6} (0.6741 + 2 \times 0.6515 + 2 \times 0.6507 + 0.6301)$$

$$\therefore y_2 = 0.2705 = y(1.2) \quad \underline{\text{Ans}}$$

2020-Fall

5. a) Apply RK-4th order method to find an approximate value of y when $x=0.3$ given that: $y' = 2.5y + e^{0.3x}$, with an initial $y(0) = 1$, taking $h=0.3$

solⁿ Here,

$$y' = 2.5y + e^{0.3x}, \quad y(0) = 1$$

$$\therefore f(x) = 2.5y + e^{0.3x}$$

$$\therefore x_0 = 0, y_0 = 1, x_n = 0.3$$

$$\therefore h = 0.3$$

The RK-4th order method is given by

$$y_{n+1} = y_n + \frac{h}{6} (m_1 + 2m_2 + 2m_3 + m_4)$$

where,

$$m_1 = f(x_n, y_n)$$

$$m_2 = f(x_n + h/2, y_n + m_1 h/2)$$

$$m_3 = f(x_n + h/2, y_n + m_2 h/2)$$

$$m_4 = f(x_n + h, y_n + m_3 h)$$

For $n=0$:

$$y_1 = y_0 + \frac{h}{6} (m_1 + 2m_2 + 2m_3 + m_4) \quad \text{--- (1)}$$

where,

$$m_1 = f(x_0, y_0) = f(0, 1) = 3.5$$

$$m_2 = f(x_0 + h/2, y_0 + m_1 h/2) = f(0.15, 1.5250) = 4.8585$$

$$m_3 = f(x_0 + h/2, y_0 + m_2 h/2) = f(0.15, 1.7288) = 5.3680$$

$$m_4 = f(x_0 + h, y_0 + m_3 h) = f(0.3, 2.6104) = 7.6202$$

From (1),

$$y_1 = \frac{y}{6} + \frac{0.3}{6} (3.5 + 2 \times 4.8585 + 2 \times 5.3680 + 7.6202)$$

$$\therefore y_1 = 2.5787 = y(0.3) \quad \underline{\text{Ans}}$$

2018-Spring

5. a) Use the Runge-Kutta 4th order method, to solve
 $10 \frac{dy}{dx} = x^2 + y^2, y(0) = 1$ for the interval $0 \leq x \leq 0.4$ with

$$h = 0.1.$$

sol Here,

$$10 \frac{dy}{dx} = x^2 + y^2, y(0) = 1$$

$$\Rightarrow \therefore \frac{dy}{dx} = \frac{x^2 + y^2}{10}$$

$$\therefore f(x) = \frac{x^2 + y^2}{10}$$

$$\therefore x_0 = 0, y_0 = 1$$

$$\therefore h = 0.1$$

The RK-4th order method is given by

$$y_{n+1} = y_n + \frac{h}{6} (m_1 + 2m_2 + 2m_3 + m_4)$$

where,

$$m_1 = f(x_n, y_n)$$

$$m_2 = f(x_n + h/2, y_n + m_1 h/2)$$

$$m_3 = f(x_n + h/2, y_n + m_2 h/2)$$

$$m_4 = f(x_n + h, y_n + m_3 h)$$

For $n=0$:

$$y_1 = y_0 + \frac{h}{6} (m_1 + 2m_2 + 2m_3 + m_4) \quad \text{--- (1)}$$

where,

$$m_1 = f(x_0, y_0) = f(0, 1) = 0.1$$

$$m_2 = f(x_0 + h/2, y_0 + m_1 h/2) = f(0.05, 1.0050) = 0.1013$$

$$m_3 = f(x_0 + h/2, y_0 + m_2 h/2) = f(0.05, 1.0051) = 0.1013$$

$$m_4 = f(x_0 + h, y_0 + m_3 h) = f(0.1, 1.0101) = 0.1030$$

from (1),

$$y_1 = 1 + \frac{0.1}{6} (0.1 + 2 \times 0.1013 + 2 \times 0.1013 + 0.1030)$$

$$\therefore y_1 = 1.0101 = y(0.1) \quad \underline{\text{Ans}}$$

For $n=1$:

$$y_2 = y_1 + \frac{h}{6} (m_1 + 2m_2 + 2m_3 + m_4) \quad \text{--- (2)}$$

where,

$$m_1 = f(x_1, y_1) = f(0.1, 1.0101) = 0.1030$$

$$m_2 = f(x_1 + h/2, y_1 + m_1 h/2) = f(0.15, 1.0153) = 0.1053$$

$$m_3 = f(x_1 + h/2, y_1 + m_2 h/2) = f(0.15, 1.0154) = 0.1054$$

$$m_4 = f(x_1 + h, y_1 + m_3 h) = f(0.2, 1.0206) = 0.1082$$

from (2),

$$y_2 = 1.0101 + \frac{0.1}{6} (0.1030 + 2 \times 0.1053 + 2 \times 0.1054 + 0.1082)$$

$$\therefore y_2 = 1.0206 = y(0.2) \quad \underline{\text{Ans}}$$

For $n=2$:

$$y_3 = y_2 + \frac{h}{6} (m_1 + 2m_2 + 2m_3 + m_4) \quad \text{--- (3)}$$

where,

$$m_1 = f(x_2, y_2) = f(0.2, 1.0206) = 0.1082$$

$$m_2 = f(x_2 + h/2, y_2 + m_1 h/2) = f(0.25, 1.0260) = 0.1115$$

$$m_3 = f(x_2 + h/2, y_2 + m_2 h/2) = f(0.25, 1.0262) = 0.1116$$

$$m_4 = f(x_2 + h, y_2 + m_3 h) = f(0.3, 1.0318) = 0.1155$$

From (3),

$$y_3 = 1.0206 + \frac{0.1}{6} (0.1082 + 2 \times 0.1115 + 2 \times 0.1116 + 0.1155)$$

$$\therefore y_3 = 1.0318 = y(0.3) \quad \underline{\text{Ans}}$$

For $n=4$:

$$y_4 = y_3 + \frac{h}{6} (m_1 + 2m_2 + 2m_3 + m_4) \quad \text{--- (4)}$$

where,

$$m_1 = f(x_3, y_3) = f(0.3, 1.0318) = 0.1155$$

$$m_2 = f(x_3 + h/2, y_3 + m_1 h/2) = f(0.35, 1.0376) = 0.1199$$

$$m_3 = f(x_3 + h/2, y_3 + m_2 h/2) = f(0.35, 1.0378) = 0.1200$$

$$m_4 = f(x_3 + h, y_3 + m_3 h) = f(0.4, 1.0438) = 0.1250$$

From (4),

$$y_4 = 1.0318 + \frac{0.1}{6} (0.1155 + 2 \times 0.1199 + 2 \times 0.1200 + 0.1250)$$

$$\therefore y_4 = 1.0438 = y(0.4) \quad \underline{\text{Ans}}$$